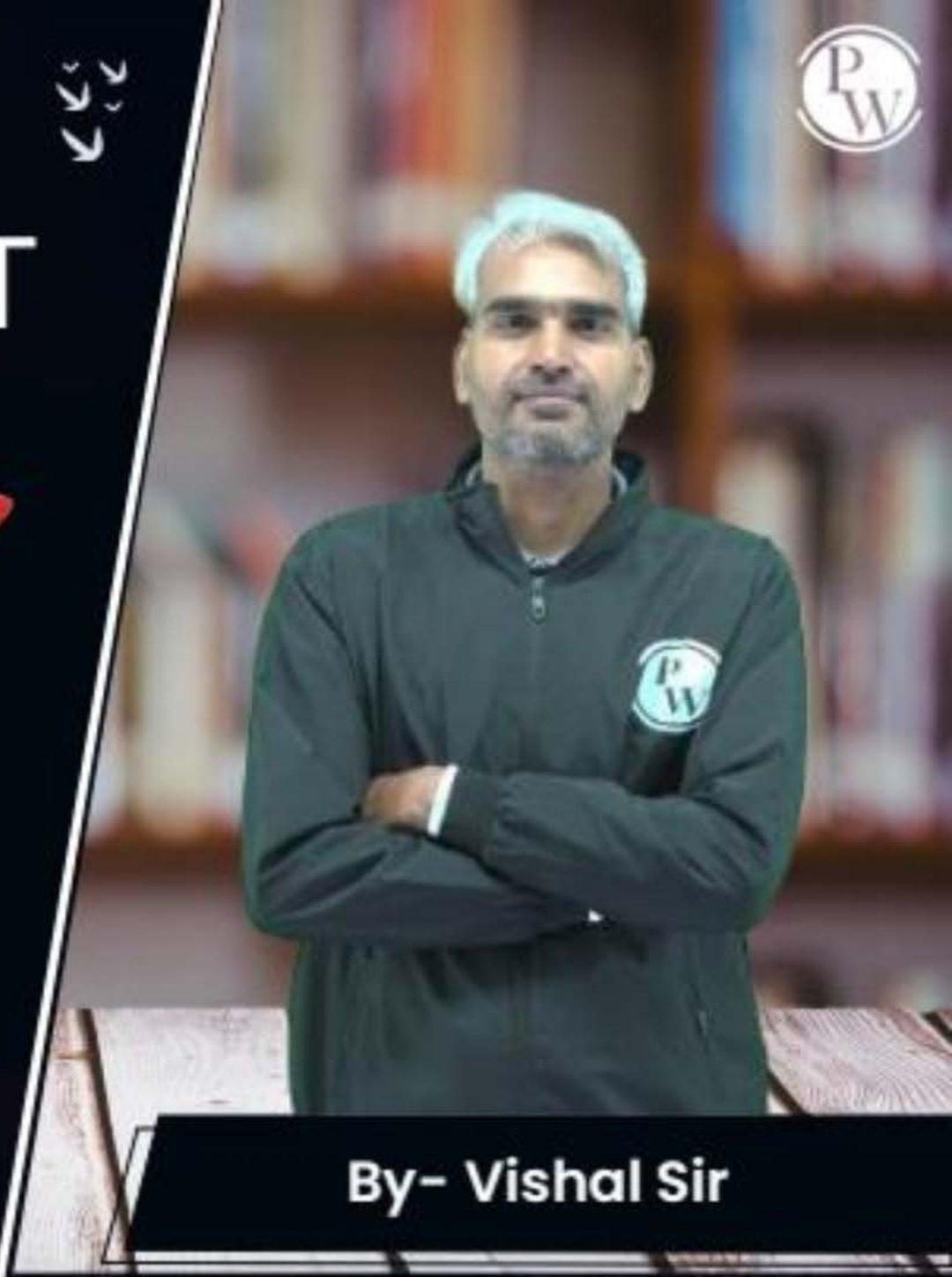
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 14



By- Vishal Sir

Recap of Previous Lecture



Topic Lossless Natural Join

Topic Normal forms

Slide

Topics to be Covered



- Topic: First normal form (1NF)
- Topic: Redundancy in relation because of FD
- Topic: Second normal form (2NF)
- Topic: Third normal form (3NF)
- Topic: Boyce codd normal form (BCNF)

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Topic : Normalization

- Normalization is the process of decomposing the relation into sub-relations, such that redundancy is reduced or eliminated.
- To specify the different types of redundancy we use the concept of "Normal Forms"



Topic : Normal forms

Boyce Codd Normal form → introduced the Concept of relational model.

There are various types of normal forms,

1 NF | 2 NF | 3NF | BCNF | 4NF

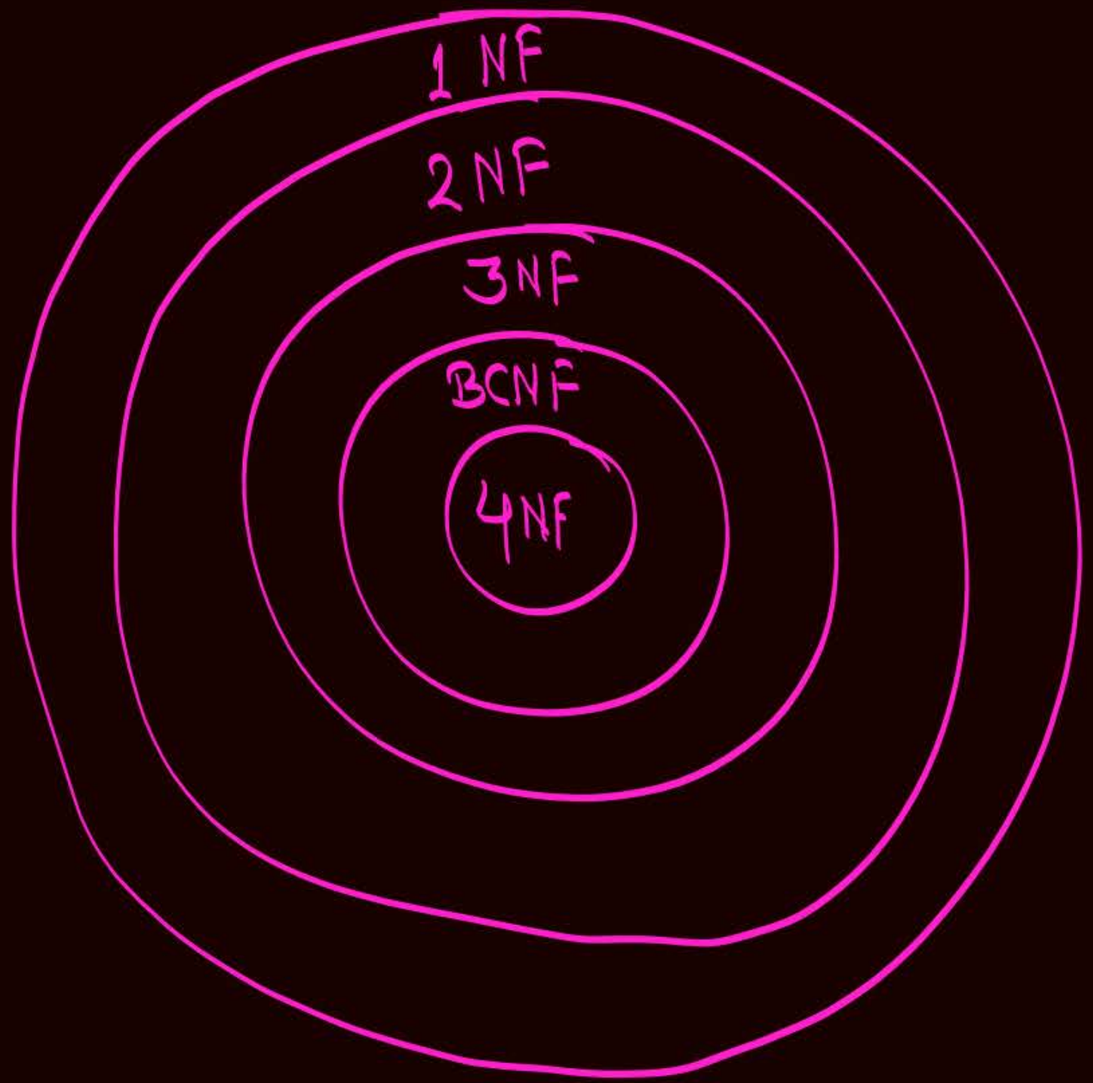
* Upto BCNF we will try to eliminate the redundancies present in the relation because of functional dependencies.
* If a relation is in BCNF, then there will be no redundancy in that relation because of functional dependencies, but a relation which is in BCNF may still suffer from redundancy present in it because of Multi-valued dependencies (MVD)

* 4NF is related to multi-valued dependencies
* In 4NF we will try to eliminate the redundancies present in the relation because of multi-valued dependency.

* Every relation which is in 2NF, is also in 1 NF.
* Every relation which is in 3NF, is also in 2NF and hence also in 1 NF

:

and so on





Topic : First normal form (1NF)

A database is in 1NF only if it does not contain any multivalued attribute {i.e. all attributes must be simple and atomic}

eg:

Sid	Mob_No.
S_1	{12345, 98270}
S_2	{94250, 98930}
S_3	98930

Multi-valued attribute

it is not a relation

Multi-valued attribute is present so Not in 1NF

Convert multi-valued attributes into single valued attributes

Sid	Mob_No.
S_1	12345
S_1	98270
S_2	94250
S_2	98930
S_3	98930

Now, Mob_No. is single valued attribute

No multi-valued attribute exists so it is atleast in 1NF.

* By default normal form of relation is 1NF.

{i.e, Every relation is at least in 1NF}



Topic : Redundancy in relation because of FD

Rule 1 :- In a functional dependency "$X \rightarrow Y$", if "X" is a Super key, then it does not cause any redundancy in the relation

Rule 2 :- In a functional dependency "$X \rightarrow Y$" if X is not a super key, then it may cause redundancy in the relation

$(A)^+ = \{A, B, C\}$

$\therefore$ A is a S.K

$A \rightarrow BC$

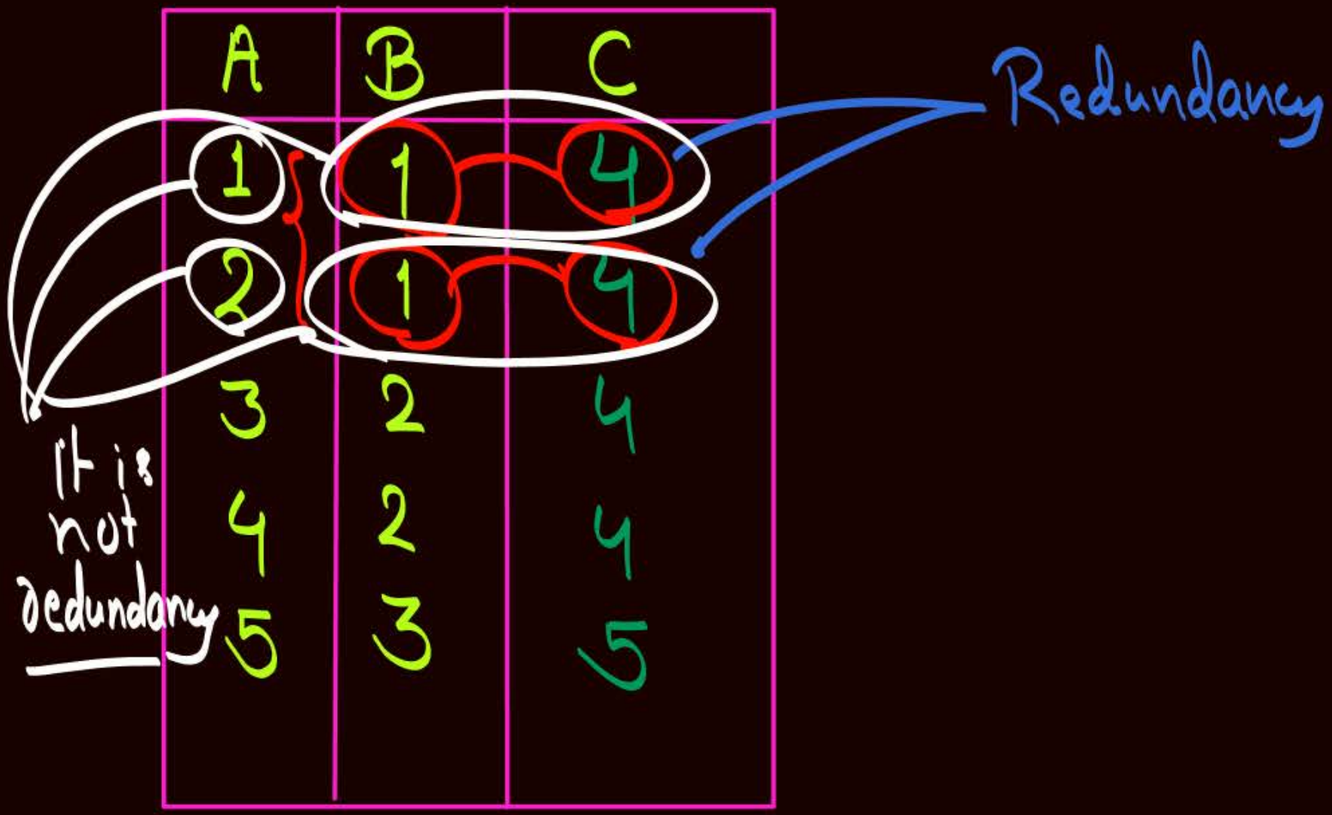
AB is a S.K

$AB \rightarrow C$

$B \rightarrow C$

$(B)^+ = \{B, C\}$

$\therefore$ Not a S.K.



Possible types of non-trivial FDs which may cause redundancy in the relation

Note:. In $X \rightarrow Y$, if X is not a Superkey, then Y can never be a Super key.

∴ If $X \rightarrow Y$ causes redundancy in the relation, then neither L.H.S. nor R.H.S. of that FD can be a Super key.

Possible types of non-trivial FDs which may cause redundancy in the relation



Type ① Proper subset of a C.K → Non-prime attributes ✓

Type ② (Proper Subset of a Candidate key + Non-prime attributes) → Non-prime attribute

Type ③ Non-prime attributes → Non-prime attributes

Non-Prime Attributes → P.S.C.K. {Such FDs are not Possible}

Proper subset of a C.K → Proper subset of same C.K Such FD is not Possible

Type ④ Proper subset of one C.K → Proper Subset of some other C.K

Type ⑤ (Proper Subset of one Candidate key + Non-prime attributes) → Proper Subset of some other C.K

Note :-

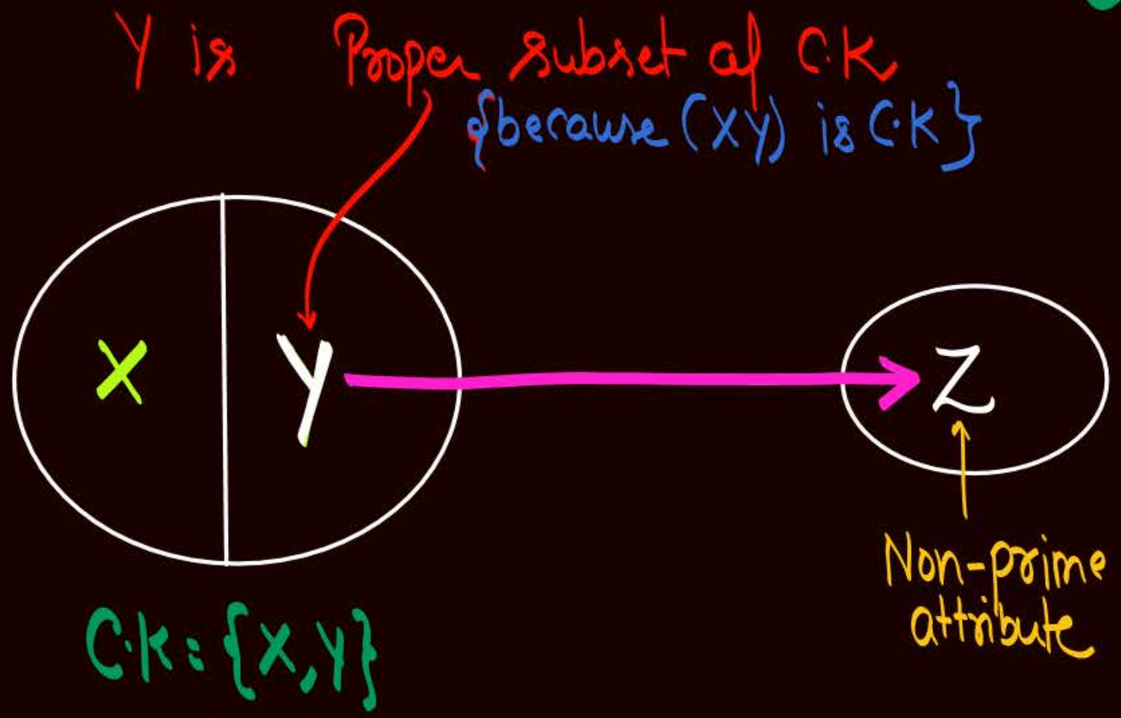
Normal Form \ FD type	Type 1	Type 2	Type 3	Type 4	Type 5
1 NF	Allowed	Allowed	Allowed	Allowed	Allowed
2 NF	Not allowed	Allowed	Allowed	Allowed	Allowed
3 NF	Not allowed	Not allowed	Not allowed	Allowed	Allowed
BCNF	Not allowed	Not allowed	Not allowed	Not allowed	Not allowed



Topic : Second normal form (2NF)

- A relation R is in 2NF if and only if,
  - ① R is in 1NF
  - and ② Relation R must not contain any Partial dependency

Partial Functional dependency



if $Y \rightarrow Z$ exists, then $XY \rightarrow Z$ is Partial functional dependency

★ Let (XY) is the candidate key and 'Z' is some non-Prime attribute in relation R

① If "Z" is fully dependent on candidate key {i.e, No proper subset of Candidate key can determine "Z"}, then "$XY \rightarrow Z$" is called full functional dependency.

② If Proper subset of Candidate key {X, Y} Can determine "Z", then $XY \rightarrow Z$ is Called Partial Functional dependency

* If type 1 functional dependency { Proper subset of C.k ⟶ Non-prime attribute } exists in the relation, then partial functional dependency exist in the relation.

∴ for 2NF,

① Relation must be in 1NF

and ② Relation must not contain any functional dependency of "type 1"



Topic : Third normal form (3NF)

A relation R is in 3NF if and only if,

① R is in 1NF {i.e, No multi-valued attribute}

and ② Every non-trivial functional dependency $X \rightarrow Y$ must have,

X as a Super key

(or)

Y as a prime attribute

→ i.e, Type-1, Type-2, and Type-3 FDs are not allowed.

In $X \rightarrow Y$, if X is a Super Key, then there will be no redundancy because of $X \rightarrow Y$ i.e $X \rightarrow Y$ will satisfy BCNF Condn and hence 3NF as well.

Note:- A relation R is in 3NF, only if transitive functional dependency does not exist in relation.

→ A functional dependency is called transitive functional dependency, if non-prime attribute is transitively (not directly) dependent on key.

C.K/S.K. ⟶ "Non-prime Attribute 1"

In this case "Non-prime attribute 1" is directly dependent on key. ∴ it is not a transitive FD ∴ Allowed in 3NF.

Neither a C.K. nor a S.K ⟶ Non-prime Attribute 2

Non-prime attribute is not directly dependent on key. ∴ Transitive FD ∴ Hence Not allowed in 3NF



Topic : Boyce codd normal form (BCNF)

* A relation R is in BCNF if and only if,
  Every non-trivial functional dependency "$X \longrightarrow Y$"
  must have "X" as a Super key.

{ None of the five types of FDs are allowed }



2 mins Summary

- Topic: First normal form (1NF)
- Topic: Redundancy in relation because of FD
- Topic: Second normal form (2NF)
- Topic: Third normal form (3NF)
- Topic: Boyce codd normal form (BCNF)

THANK - YOU

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